Md Nahidul Islam

mislam19@memphis.edu | +1 (901) 319-6770 | Memphis, TN, USA
personal website | Google Scholar | ORCID | DBLP | GitHub | LinkedIn

Research Interests

Autonomous driving, interactive motion planning, trajectory forecasting, trustworthy machine learning, conformal risk certification, counterfactual evaluation, model compression, knowledge distillation, and safe robotics.

Research Profile: Ph.D. researcher building scalable evaluation and certification methods for autonomous-driving decision systems, with experience in CARLA, trajectory datasets, PyTorch, and HPC/GPU experimentation.

Education

Doctor of Philosophy (Ph.D.) in Computer Science — *2024 – Present*
UNIVERSITY OF MEMPHIS — Memphis, TN, USA
Advisor: Dr. Myounggyu Won | GPA: 3.93/4.00 | M.S. earned en route, May 2026
M.S. Thesis: BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning

Bachelor of Science (B.Sc.) in Computer Science and Engineering — *2012 – 2017*
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY (BUET) — Dhaka, Bangladesh

Publications and Manuscripts

Full list available at: Google Scholar | ORCID | DBLP

*Author of this CV in **bold**. Submitted manuscripts are clearly marked as under review.*

[1] **M. N. Islam**, et al., "CRiC: Protocol Scoped Counterfactual Risk Certification for Interactive Driving," *submitted to NeurIPS 2026.*

[2] **M. N. Islam**, et al., "BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning," *submitted to IROS 2026.*

[3] **M. N. Islam**, "BucketKD: A Safety-Aware Bucket-Based Knowledge Distillation Framework for End-to-End Motion Planning," M.S. thesis, Dept. of Computer Science, University of Memphis, Memphis, TN, USA, 2026. [link]

[4] L. Hwang, **M. N. Islam**, M. H. Ali, D. Dasgupta, and M. Won, "Detecting Stealthy Anomalies with Autoencoders Using Windowed Variance-Aware Loss Functions," in *Proc. IEEE Military Communications Conference (MILCOM)*, 2025, pp. 981–986. DOI: 10.1109/MILCOM64451.2025.11310014.

[5] N. Sadeq, S. Ahmed, S. S. Shubha, **M. N. Islam**, and M. A. Adnan, "Bangla Voice Command Recognition in End-to-End System Using Topic Modeling Based Contextual Rescoring," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 7894–7898.

[6] S. Ahmed, N. Sadeq, S. S. Shubha, **M. N. Islam**, and M. A. Adnan, "Preparation of Bangla Speech Corpus from Publicly Available Audio & Text," in *Proc. 12th Int. Conf. Language Resources and Evaluation (LREC)*, 2020, pp. 6586–6592.

[7] S. S. Shubha, N. Sadeq, S. Ahmed, **M. N. Islam**, M. A. Adnan, M. Y. A. Khan, and M. Z. Islam, "Customizing Grapheme-to-Phoneme System for Non-Trivial Transcription Problems in Bangla Language," in *Proc. NAACL-HLT*, 2019, pp. 3192–3201. *[ACL Travel Grant Award.]*

Research Experience

Graduate Research Assistant — *Jan. 2024 – Present*
UNIVERSITY OF MEMPHIS – ADVISOR: PROF. MYOUNGGYU WON — Memphis, TN, USA

- Designed *BucketKD*, a safety-aware bucket-based knowledge-distillation framework for compressing end-to-end motion planners in autonomous driving; evaluated compact student planners in CARLA high-complexity driving scenarios.
- Developed time-to-collision (TTC)-guided waypoint attention and adaptive planning-state discretization to preserve safety-relevant trajectory behavior during distillation.
- Investigating protocol-scoped counterfactual risk certification for interactive autonomous driving, including frozen decision protocols, calibrated upper-risk bounds, abstention-aware selection, and scene-level reliability evaluation.
- Built scalable experimental pipelines for autonomous-driving research using PyTorch, CARLA, Waymo Open Motion Dataset (WOMD), Argoverse 2 (AV2), highD, inD, rounD, JSONL artifacts, and HPC/GPU infrastructure.
- Co-developed an autoencoder-based stealthy-anomaly detector with a windowed variance-aware loss for vehicular/tactical communication networks, published at IEEE MILCOM 2025.

Research Assistant *Aug. 2018 – Jan. 2020*
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY (BUET) Dhaka, Bangladesh

- Conducted research on Bangla speech recognition, low-resource language modeling, and grapheme-to-phoneme conversion.
- Contributed to peer-reviewed publications at ICASSP, LREC, and NAACL.
- Built and evaluated sequence-to-sequence and attention-based models for Bangla ASR and text/speech processing.

Selected Projects

CRiC: Counterfactual Risk Certification for Interactive Driving *Ongoing*
PH.D. RESEARCH PROJECT | PYTORCH, TRAJECTORY FORECASTING/PLANNING, CONFORMAL CALIBRATION, HPC

- Developed a protocol-scoped certification framework for interactive autonomous driving that evaluates a frozen decision protocol rather than only a predictor.
- Designed calibrated upper-risk-bound selection and abstention logic for safety-aware candidate choice under uncertainty.
- Built streaming-friendly evaluation artifacts and reliability analyses for large-scale scene/candidate rollouts.

BucketKD: Safety-Aware End-to-End Motion Planner Compression *Jan. 2024 – Feb. 2026*
M.S. THESIS, UNIVERSITY OF MEMPHIS | PYTORCH, CARLA SIMULATOR

- Developed a bucket-based knowledge-distillation framework for compressing end-to-end autonomous-driving planners while preserving safety-relevant planning behavior.
- Designed TTC-based waypoint attention and adaptive bucket construction to improve safety-aware distillation.

Autonomous R/C Cars – Cybersecurity Summer Camp *Summer 2024*
DONKEY CAR, RASPBERRY PI, PYTHON, TENSORFLOW, OPENCV

- Built an end-to-end self-driving pipeline – data collection, CNN training, on-vehicle deployment – on Raspberry Pi-based R/C platforms for a five-day high-school camp (associated with SIGCSE 2024).
- Designed adversarial demonstrations including MITM, password cracking, and buffer-overflow exploits to teach safety-critical attack surfaces.

Technical Skills

Programming Python, C++, C, C#, Java, JavaScript, SQL/PL-SQL, R

Machine Learning PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, pandas, NumPy, spaCy, NLTK

Autonomous Driving / Robotics CARLA, Donkey Car, trajectory forecasting, motion planning, TTC-based safety metrics, Waymo Open Motion Dataset (WOMD), Argoverse 2 (AV2)

Systems / Infrastructure Linux, Git, Docker, Kubernetes, HPC/GPU clusters, AWS, DigitalOcean, CI/CD, Ansible

Web / Backend Django, FastAPI, React, Redux, Node.js, ASP.NET

Databases Oracle, PostgreSQL, MySQL, MS SQL Server

Industry Experience

Assistant Director, Enterprise Data Warehouse *Nov. 2020 – Dec. 2023*
BANGLADESH BANK (CENTRAL BANK OF BANGLADESH) Dhaka, Bangladesh

- Designed and maintained ETL pipelines and internal analytics platforms for enterprise-scale regulatory financial data.
- Developed containerized web applications and proof-of-concept ML tools using Django, React, Oracle, Docker, and CI/CD workflows.

Software Engineer, AI Solutions Team *Feb. 2020 – Oct. 2020*
BINATE SOLUTIONS LTD. Dhaka, Bangladesh

- Developed REST APIs, ML-backed services, and SaaS web applications using Django, TensorFlow/Rasa, and cloud deployment tools.

Information Analyst, MIS Department *Feb. 2017 – Mar. 2018*
ACI LTD. Dhaka, Bangladesh

- Built reporting dashboards and applied machine-learning methods to business and operations datasets.

Teaching Experience

Graduate Teaching Assistant *Jan. 2024 – Present*
UNIVERSITY OF MEMPHIS, DEPARTMENT OF COMPUTER SCIENCE Memphis, TN, USA

- Mentored *Capstone Software Project* and *Operating Systems* courses of 50–80 students; evaluated projects, and proctored exams.